

OPIM 5512 • LAB 2

Why SHAP is fair: split the prize 🏆

From a Kaggle prize → to a prediction. The weights are just counting.

WARM-UP • 2 PLAYERS

You + a friend win a \$10,000 prize 

1/2, 1/2

A time machine lets each of you redo the contest **alone**: you place 2nd (7,500), your friend 3rd (5,000), nobody → 0. Every way to form the team — the **prize** $v(S)$:

team	{ }	{You}	{Friend}	{You, Friend}
prize $v(S)$	0	7,500	5,000	10,000

2 players → 2! = 2 orderings → each bump counts 1/2 (a plain average):

- **You:** join nobody (+7,500) & join friend, 3rd→1st (+5,000) → $\frac{1}{2}(7,500) + \frac{1}{2}(5,000) = \mathbf{6,250}$
- **Friend:** +5,000 and +2,500 → **3,750** $6,250 + 3,750 = \mathbf{10,000}$ ✓

WARM-UP • 3 PLAYERS

Add a teammate → 8 teams, 3! = 6 orderings

$\frac{1}{3}, \frac{1}{6}, \frac{1}{6}, \frac{1}{3}$

team	{ }	P1	P2	P3	P1,P2	P1,P3	P2,P3	all
prize v(S)	0	5,000	5,000	0	7,500	7,500	5,000	10,000

With 3 players there are 3! = 6 ways to form the team, so the bumps count $\frac{1}{3}, \frac{1}{6}, \frac{1}{6}, \frac{1}{3}$.

Fair shares: **P1 = 5,000 P2 = 3,750 P3 = 1,250** → sum **10,000** ✓

(both examples in USD, from Conor O’Sullivan, “From Shapley to SHAP”)

THE KEY IDEA

The weights are **FIXED** — just $1/n!$ counting 

The weights depend **only on how many players there are** — never on the data:

2 players → $2! = 2$ orderings → $\frac{1}{2}, \frac{1}{2}$

3 players → $3! = 6$ orderings → $\frac{1}{3}, \frac{1}{6}, \frac{1}{6}, \frac{1}{3}$

With 3 things there are only **6 ways to form the team**, so the weights are **locked in** — the same $\frac{1}{3}, \frac{1}{6}, \frac{1}{6}, \frac{1}{3}$ for *every* problem, every dataset.

Only the $v(S)$ scores change from row to row.

NOW THE MODEL

It's the 3-player game



The **players are the features** (Income, Rooms, Latitude) and the **prize is the prediction**.

One twist: a model doesn't start at **0** — it starts at the **baseline prediction** (here 10). So the fair shares sum to *prediction – baseline* and you add the baseline back:

$$\text{base } 10 + (\varphi_I=8 + \varphi_R=2 + \varphi_L=4) = 24$$

Same fair-split machinery. Same $\frac{1}{3}$, $\frac{1}{6}$, $\frac{1}{6}$, $\frac{1}{3}$ weights. Now let's trace the tree.